

MATRIMORPHOSIS – 2026

ASSIGNMENT 3

INTRODUCTION TO LINUX

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1. Is it possible to create a hard link to a directory in Linux? If not, why?

- No, it is generally not possible for a normal user to create a hard link to a directory in Linux. A hard link is a directory entry that refers to the same inode as another entry. Hard links are normally allowed for regular files, but Linux restricts creating hard links to directories.
- The main reason is that allowing directory hard links could create circular structures in the filesystem. For example, one directory could point to another directory that eventually points back to the first directory. This could create cycles and cause problems while traversing and maintaining the filesystem.
- Command to test:
 - `mkdir testdir`
 - `ln testdir hardlink`
- Expected result: The command should produce an error indicating that a hard link to a directory is not allowed.

2. Is it possible to create a file or a directory in Linux with a name that contains a space? For example, how do you create a directory named My Documents?

- Yes, it is possible to create both files and directories with spaces in their names in Linux. However, the space has a special meaning in the Linux shell because it is normally used to separate different arguments in a command. Therefore, when a filename or directory name contains spaces, the complete name must be treated as a single argument.
- There are two common ways to handle spaces in Linux. The first is to enclose the complete name within quotation marks. The second is to use a backslash (\) before the space.
- Command – Method 1 (using quotation marks):
 - `mkdir "My Documents"`
 - This creates a directory with the name My Documents.

- Command – Method 2 (using backslash):
 - `mkdir My\ Documents`
 - This also creates a directory with the name My Documents.
- Command to verify:
 - `ls`
 - The directory My Documents should appear in the listing.
- Example for creating a file with spaces:
 - `touch "My File.txt"`
 - This creates a file named My File.txt.

3. In your current working directory, you want to create a file and a directory, both with the same name. For example, let us call them both by the name sample. Is it possible to create both? Try it out yourself by using the touch and mkdir commands. How do you explain the behaviour?

- No, it is not possible to create both a file and a directory with exactly the same name in the same directory. In Linux, each directory contains entries identified by their names, and two entries in the same directory cannot have the same name, even if one is a file and the other is a directory.
- The touch command is used to create a new empty file if a file with the specified name does not already exist. The mkdir command is used to create a new directory with the specified name. If an entry with that name already exists, the command will fail.
- Command – First create a file
 - `touch sample`
- Now try to create a directory with the same name:
 - `mkdir sample`
- Expected result:
 - The mkdir command will display an error because sample already exists.

- Command – First create a directory
 - If you remove the previous sample file, you can test the opposite order:


```
mkdir sample
```
 - Now try to create a file with the same name:


```
touch sample
```
 - Expected result:

The touch command does not create a new file because sample already exists as a directory. Instead, it updates the access and modification timestamps of the existing directory.
- Explanation of the behaviour
 - The behaviour occurs because a directory cannot contain two entries with the same name. When sample is already a file, mkdir sample cannot create a directory with the same name. When sample is already a directory, touch sample does not create another file; it treats the existing directory as an existing filesystem object and can update its timestamps.
- Command to verify
 - `ls -l`
 - This will show whether sample is currently a file or a directory.

4. What is the ls command (with options) equivalent to the dir command?

- The dir command is commonly used in Windows to display the files and directories present in the current directory. In Linux, the equivalent command for listing files and directories is ls.
- The ls command can be used with different options to display additional information about files and directories.
- Command – Basic listing:
 - `ls`
 - This displays the names of files and directories in the current working directory.
- Command – Detailed listing:
 - `ls -l`

- The `-l` option displays a detailed listing that includes information such as file permissions, number of links, owner, group, file size, and modification date and time.
- Command – Listing including hidden files:
 - `ls -la`
 - The `-a` option displays hidden files and directories along with the other files.
- Therefore, the basic equivalent of the Windows `dir` command in Linux is:
 - `dir` → `ls`
- The options can be added to `ls` when more detailed information or hidden files need to be displayed.